2. $F = \overline{A(\bar{A}B+C)} + (A+B)(\bar{A}+C) + AC\,\overline{(A\bar{B})} + \overline{(\bar{A}+\bar{B}+C)}$

$A\overline{(\bar{A}B+C)} = A\overline{(\bar{A}B)}\cdot\bar{C} = A(A+\bar{B})\bar{C} = A\bar{C}$

$(A+B)(\bar{A}+C) = A\bar{A} + AC + \bar{A}B + BC = AC + \bar{A}B + BC$

$AC\overline{(A\bar{B})} = AC(\bar{A}+B) = AC\bar{A} + ABC = ABC$

$\overline{(\bar{A}+\bar{B}+\bar{C})} = AB\bar{C}$

$F = A\bar{C} + AC + \bar{A}B + BC + ABC + AB\bar{C}$

$= A + \bar{A}B + BC + ABC + AB\bar{C} = A + \bar{A}B + BC = \boxed{A+B}$

3.

